Name: _____ Date: _____ Period: _____

Integrity Statement:

I acknowledge that I will complete this practice assessment using only my own work. Show all reasoning for free-response questions.

Unit 1 Quiz 1 - Kinematics in One Dimension

Mock Exam C · Form 2: Connecting ideas

Multiple Choice Questions: Select the best answer. (1 point each)

1. A ball is thrown vertically upward from the ground at 19.6 m/s. Ignore air resistance and use g = 9.8 m/s². Find the maximum height above ground.

(A) 39.2 m

(B) 9.8 m

(C) 19.6 m

(D) -19.6 m

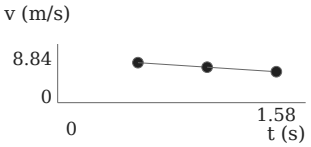
2. A bicycle moving right brakes uniformly. Use the model only until it stops. t (s) -> v (m/s): (0.58, 8.84); (1.08, 7.84); (1.58, 6.84). Which horizontal setting produces a reading of 8.34?

(A) 1.162

(B) 0.498

(C) 1.494

(D) 0.83



3. A student walks 10 m right from a sensor, then 3 m left in a total of 6 s. Taking right as positive, which pair gives average speed and average velocity?

(A) Both zero

(B) 1.167 m/s; 2.167 m/s

(C) 2.167 m/s; +1.167 m/s

(D) 7 m/s; 13 m/s

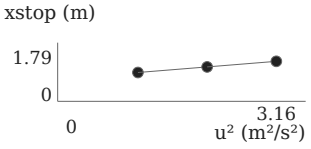
4. A cart brakes immediately at a constant magnitude of 2 m/s². Read its stopping coordinate along the track. u² (m²/s²) -> xstop (m): (1.16, 1.29); (2.16, 1.54); (3.16, 1.79). Set the horizontal origin at the first measurement. What is the new vertical intercept?

(A) 2.322

(B) 1.29

(C) 1.806

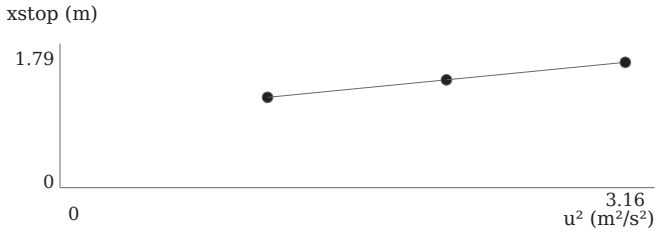
(D) 0.774



Free Response Questions: Answer all parts. Show your work and clearly indicate your answers.

5. A cart brakes immediately at a constant magnitude of 2 m/s². Read its stopping coordinate along the track. u² (m²/s²) -> xstop (m): (1.16, 1.29); (2.16, 1.54); (3.16, 1.79).

a. For the stopping-distance test setup, label the measured quantities and their reference directions or origins. **(1 pt)**



b. Exchange the axes. Relate the new slope and its units to the original slope. **(3 pt)**

6. A cart initially moving at -2 m/s receives a constant acceleration lasting 2 s each trial. a (m/s²) -> vfinal (m/s): (1.16, 0.32); (2.16, 2.32); (3.16, 4.32).

a. Build the model for velocity-area experiment from physics and predict both slope and intercept. **(2 pt)**

b. Exchange the axes. Relate the new slope and its units to the original slope. **(2 pt)**

7. A bicycle moving right brakes uniformly. Use the model only until it stops. t (s) -> v (m/s): (0.58, 8.84); (1.08, 7.84); (1.58, 6.84). Build the model for brake recorder from physics and predict both slope and intercept. **(3 pt)**